SIMATS ENGINEERING
Department of Computer Science and Engineering
ASSESSMENT TOOL 2 – CONCEPT MAPPING

Course Code:	CSA1220	**Course Title:**	Computer Architecture
CO Assessed:	CO1 – Precision (BL4)	**Assessment:**	Concept Mapping
Weightage:	35%	**Total Marks:**	100
CO1 Statement:	Analyse the functional units of a computer system including CPU, memory, bus structures, and I/O components by examining instruction execution cycles, addressing modes, and performance metrics such as CPI, clock cycles, and benchmarking		
Student Name:	T. Rishitha	**Reg. No.:**	192472396
Section / Batch:	2024	**Date:**	17/07/26

Instructions to Students

- Identify the key computer architecture concepts, components, and performance parameters mentioned in the given topic or scenario.
- Organize the identified concepts logically by establishing relationships between CPU, memory, I/O devices, bus structures, instruction execution cycle, and performance metrics.
- Represent the concepts using an appropriate concept map with clear nodes, connecting lines, and meaningful linking phrases.
- Demonstrate the hierarchy and interdependence among concepts, showing how one concept influences or interacts with another.
- Ensure the concept map is complete, accurate, well-structured, and easy to understand by using appropriate terminology and logical connections.

QUESTIONS

1. A computer manufacturer is designing a workstation for engineering simulations that frequently exchanges data between the processor, memory, and peripheral devices. Design a concept map illustrating the interaction among the CPU, Memory Unit, Input Unit, Output Unit, and I/O devices, showing how instructions and data move through the system during program execution. How do the functional units of a computer system cooperate to execute instructions efficiently?

2. A software developer notices that an application takes longer to execute because several instructions require multiple memory accesses. Develop a concept map showing how the Fetch, Decode, and Execute stages of the instruction cycle are related to memory access, instruction execution, and control flow. How does each stage of the instruction execution cycle contribute to the successful execution of a program?

3. A data center experiences communication delays because the processor, memory, and I/O devices frequently compete for access to the system bus. Prepare a concept map demonstrating the relationships among Single-bus, Multi-bus, and Hierarchical Bus organizations and their connections to the CPU, Memory, and I/O devices. How do different bus organizations improve communication efficiency and reduce system bottlenecks?

4. A microprocessor programmer is developing software for both 8085 and 8086 systems and must understand how instructions access data stored in memory. Represent the relationships among the instruction sets, register organization, addressing modes, memory organization, and memory segmentation (8086) using a concept map. How do the architectural differences between the 8085 and 8086 affect instruction execution and memory management?

5. A firmware engineer is examining machine code stored in memory while debugging an embedded application. Create a concept map explaining the relationships between Binary and Hexadecimal Number Systems, instruction representation, memory storage, data conversion, and CPU interpretation. How do binary and hexadecimal representations simplify instruction execution, memory organization, and debugging in computer systems?

Code of Conduct [192472396]

I T. Rishitha *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: T. Rishitha

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand

Marks Evaluation:

Q. No	Concept Identification (25)	Linkages (25)	Organization (20)	Integration of Literature (20)	Clarity (10)	Total (out of 100)
1	/ 4	/ 4	/ 4	/ 4	/ 4	
2	/ 4	/ 4	/ 4	/ 4	/ 4	
3	/ 4	/ 4	/ 4	/ 4	/ 4	
4	/ 4	/ 4	/ 4	/ 4	/ 4	
5	/ 4	/ 4	/ 4	/ 4	/ 4	
	/ 25	/25	/ 20	/ 20	/ 10	/ 100

Course Coordinator **Faculty Incharge**

V. Kishitha

COL-AT 2

Reg.No : 192472596

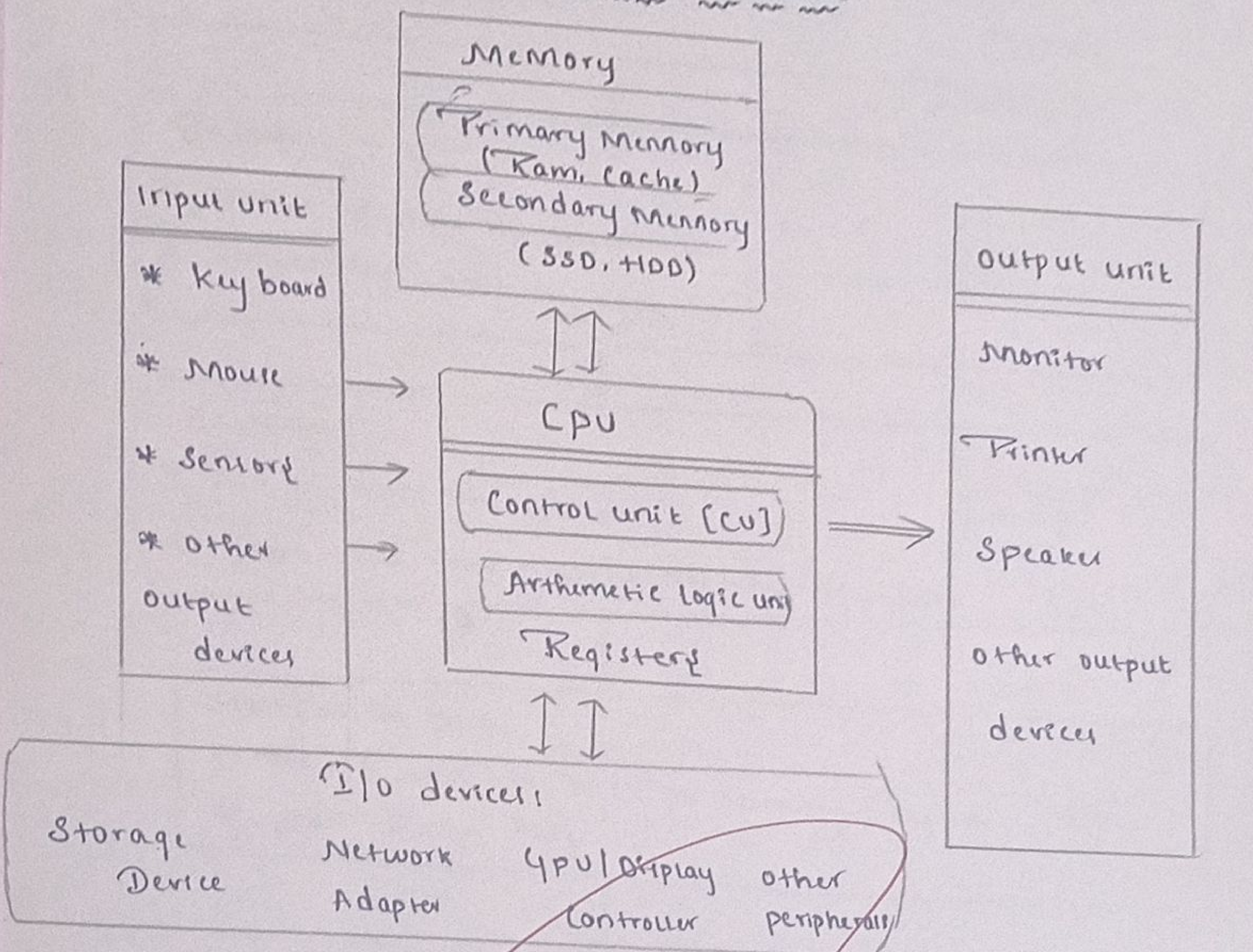
Course code : CSA1220

Course Name: Computer Architecture

1) COL- Assessment Tool - 2

Work Station for Engineering Simulations

CPU- Memory-I/O Interaction



Functional unit Cooperation

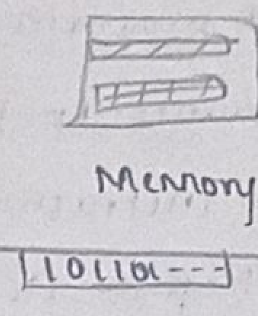
* CU coordinates all operations * ALU performs Calculation * Register Store temporary data * I/O devices Exchanges with System.

Instruction Execution Cycle

fetch → Decode → Execute

fetch:

* Cpu fetches Instruction from memory
* Instruction copied to IR



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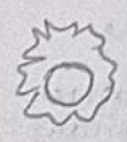
Decode:

* Control decodes the Instruction
* Determines operation type



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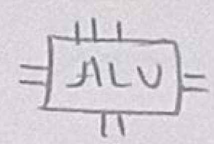
Execute:

* ALU performs the operation
* Data read from register / memory if needed



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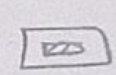
Store / write back:

* Result written back to memory / Register
* update status flag



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Control flow:

* Check for next Instruction Address
* Repeat cycle

Continious cycle until program Ends

Components Involved
* program Counter (pc)
* Instruction Register IR
* CU
* ALU
* Register
* Memory

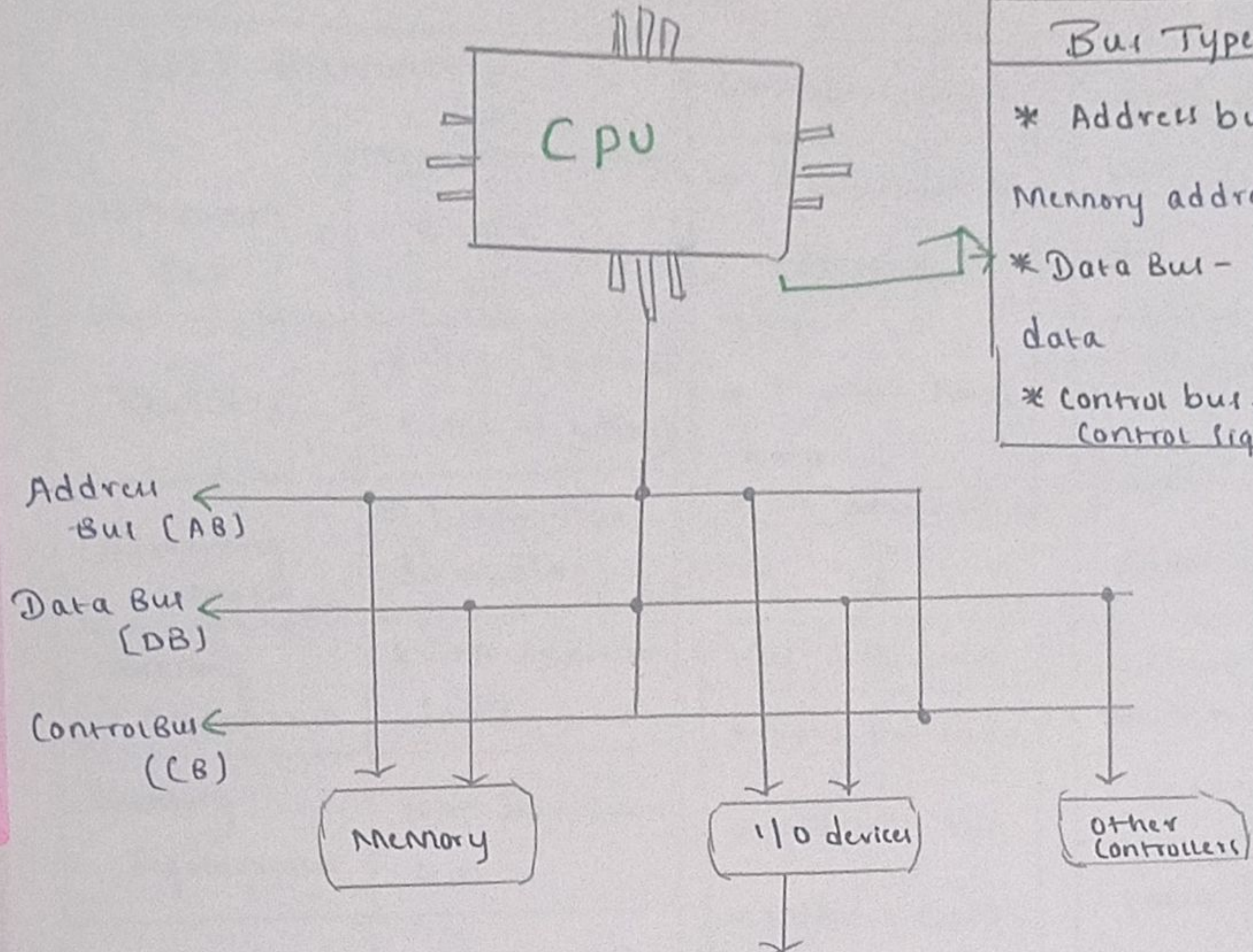
How Each Stage Contributed
fetch → get Instruction
Decode → under stand
Execute → perform operations
Store
Control

Communication in data center

Bus organizations and connections

Bus Types

* Address bus – Carries Memory addresses
* Data Bus – Carries data
* Control bus – Carries Control signals



* Network Interface
* Storage devices
* I/O Controllers
* Sensors/Acutators

Bus organizations

* Single Bus
* Multi-Bus
* Heirachial – Bus.

Eg: [CPU Memory bus | I/O buses]

Benefits

* Improves data transfer Efficiency
* Supports Multi devices Simultaneously
* Reduces comunication delays
* Minimize System bottle necks

8085 vs 8086 - Instruction Access & Memory Management

8085 Architecture

Instruction Set	8-bit
Registers	8-bit (eg: A, B, C, D, E, H, L flags)
Addressing Modes	5 (Immediate, direct, flag)
Memory Organization	64 KB Address space
Memory Segmentation	No-Segmentation
Instruction Execution	8 bit data processed at a time

Common Concepts

* fetch → decode → Execute
* cycle
* Register stores data
* I/O Communication

8086 Architecture

- Instruction Set
- Register
- Addressing Modes
- Memory Organization
- Memory Segmentation
- Instruction Execution.

Key differences

* Data bus width
* Address Bus
* Memory Capacity
* Segmentation
* performance

Impact of Memory Management & Execution.

* 8086 can handle Larger programs & data * Multitasking

Binary Vs Hexadecimal Representation & Advantages:

Binary Number System

Base: 2 (0, 1)

Example: 101101_2

→

Comparision

4 bits (binary) = 1 digit (hex)

Ex: $1011\ 1100_2$ $= BC_{16}$

Hex compact rep of binary

Hexa decimal Num System

Base: 16 (0-9, A-F)

Ex: $2FA_{16}$

Used in computer systems internal representation

- Logic circuits
- Cell states
- Machine code

Advantages

* Matches digital circuits
* fast processing
* No conversion needs inside CPU

Used by programmers (Readable Representation)

- Binary
- Write
- Debugging

Advantages

* Shorter & Easier to read
* Converts Easily to form binary

Applications

* Machine code
* Memory Addressing
* Embedded Systems

Why They both Matter?

Binary → Execution

Hex → debugging